School of Computer Science Engineering and Information Systems

5Yr Integrated M.Tech(Software Engineering)

SWE1014 : Enterprise Resource Planning

WINSEM 2024-2025

Embedded Course Project Report

TEXTILE MANAGEMENT SYSTEM

Submitted for the Course SWE1014 : Enterprise Resource Planning

Offered by Dr. Sumathi.D during WINSEM 2024-2025

By

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Vellore Institute of Technology

(Deemed to be University under section 3 of UGC Act, 1956)

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A Report on the Course Project

TEXTILE MANAGEMENT SYSTEM

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Project Title: TEXTILE MANAGEMENT SYSTEM

1. Problem Statement

1.1 Background (System Study Details in brief)

The textile industry plays a huge role in our daily lives—not just in how we dress, but also in how we express ourselves and even how we're perceived. As this industry continues to grow, so does the complexity of managing it. From receiving orders and handling inventory to managing employees and delivering finished goods, there's a lot happening behind the scenes.

Currently, many of these operations are handled manually, which can be time-consuming, inefficient, and prone to errors. That's where the **Textile Management System (TMS)** comes in. This project is designed to bring everything under one roof using automation. With the help of Odoo (an open-source ERP platform), we aim to simplify and streamline the entire process—from production to delivery—so textile businesses can focus more on quality and growth, and less on paperwork.

1.2Problem Statement

Managing a textile business manually is no small feat. Employees often have to juggle multiple tasks like order processing, material tracking, salary calculations, customer billing, and more without a centralized system.

This can lead to:

- Extra workload and stress on staff
- Delays in processing and decision-making
- Errors in inventory, payroll, and production data
- Poor coordination between different departments
- A general lack of visibility into the overall operations

1.3Novelty

- **All-in-One ERP Integration:** We're using Odoo ERP, a powerful and modern tool that brings everything from sales to HR into one connected platform.
- **User-Friendly Design:** The system is designed to be intuitive and easy to use, even for people who aren't tech-savvy.
- **Real-Time Monitoring:** Managers can keep track of what's happening across the business at any time with live dashboards and reports.
- **Better Decision Making:** With all data available in one place, business owners can make faster, more informed decisions.
- **Customizable and Scalable:** The system can grow and evolve with the business, and can easily be tailored to meet specific needs.

2. Related Works

2.1 Literature Survey (Should be elaborately discussed with its citation)

1. **Evaluating the Key Determinants of ERP Implementation Success in Garment Industries**
   Pradipta Jana and Ajay Jain (2023) conducted a study involving 65 garment businesses to identify critical success factors for ERP implementation. They found that effective leadership, clear objectives, active user participation, strategic planning, and comprehensive training are pivotal for successful ERP adoption. The study emphasizes the necessity of performance monitoring throughout the implementation process.

2. **Enhancing ERP Responsiveness Through Big Data Technologies: An Empirical Investigation**
   Florie Bandara et al. (2023) explored how integrating Big Data technologies can enhance ERP system responsiveness. Their research indicates that leveraging Big Data analytics allows organizations to process and analyze vast amounts of data in real-time, leading to improved decision-making and operational efficiency. The study highlights the importance of data integration for achieving a competitive edge.

3. **Brijraj Fashion: Style, Strength, and Success with Odoo**
   Brijraj Fashion Pvt Ltd, a leading women's ethnic wear brand in India, implemented Odoo ERP to streamline their operations. Facing challenges with data consistency and reporting, they adopted Odoo's integrated modules, resulting in a 50% time savings in managing bookkeeping and accounting data. The case study demonstrates the effectiveness of Odoo in unifying various business processes.

4. **Industry 4.0 in Textile and Apparel Industry: A Systematic Literature Review and Bibliometric Analysis of Global Research Trends**
   B. Deepthi and Vikram Bansal (2024) conducted a comprehensive review of Industry 4.0 technologies in the textile sector. Their analysis reveals a growing trend towards adopting smart manufacturing practices, including IoT and automation, to enhance productivity and flexibility in textile production. The study provides insights into global research trends and the future direction of Industry 4.0 in textiles**.**

5. **Odoo's Transformation: Elevating Sportswear Manufacturing at TIE**
   TIE, a sportswear manufacturer in Egypt, transitioned to Odoo's open-source ERP platform to overcome challenges with their outdated system. The implementation led to streamlined integration with third-party applications, operational efficiency, and significant cost reductions. This case underscores the adaptability of Odoo in the manufacturing sector.

6. **A Comprehensive Approach to Study the Adoption and Implementation of Cloud-Based ERP Among SMEs**
   Researchers (2023) examined the factors influencing SMEs' adoption of cloud-based ERP systems. Utilizing the UTAUT2 model, the study identified that budget constraints, knowledge, management support, and organizational culture significantly affect adoption decisions. The findings assist SMEs in understanding the critical elements for successful cloud ERP implementation.

7. **Use of Big Data for Textile Fashion and Garment Industry**
   N. Tarafder (2024) discussed the application of Big Data analytics in the textile and garment industry. The study highlights how Big Data can transform various aspects of the industry, including supply chain management, customer preferences analysis, and trend forecasting, leading to more informed decision-making and strategic planning.

8. **Industrial Big Data Analytics: Challenges, Methodologies, and Applications**
   JunPing Wang et al. (2018) provided an overview of the challenges and methodologies in industrial Big Data analytics. Although not specific to textiles, the study's insights into data ingestion, management, and analytics are applicable to the textile industry's move towards data-driven operations.

9. **Exploring Cloud Enterprise Resource Planning and Open Innovation for Small and Medium Enterprises: Insights from Practitioners**
   Researchers (2024) investigated how cloud ERP systems can enhance organizational performance in SMEs. The study found that with the right resources, capabilities, and culture, SMEs can leverage cloud ERP for increased agility and innovation, promoting external collaboration and reducing costs.

10. **Big Data Analytics-Enabled Dynamic Capabilities and Market Performance: Examining the Roles of Marketing Ambidexterity and Competitor Pressure**
    Gulfam Haider et al. (2024) explored the impact of Big Data analytics on marketing strategies and market performance in Pakistan's textile sector. The study suggests that Big Data capabilities can enhance marketing ambidexterity, allowing firms to better navigate market dynamics and improve performance.

2.2Comparative statement (10latest Journal papers in the current domain, Tabulation only)

S. No	Title & Author(s)	Year	Methodology	Tool / Technology Used	Advantage	Disadvantage
1	Evaluating the Key Determinants of ERP Implementation Success in Garment Industries by Pradipta Jana, Ajay Jain	2023	Survey of 65 garment SMEs, Factor Analysis	SPSS, Structured Questionnaire	Highlights key ERP success factors (training, leadership, planning)	Focused only on SMEs in India
2	Enhancing ERP Responsiveness Through Big Data Technologies by Florie Bandara et al.	2023	Empirical investigation with Big Data-ERP framework	Big Data Analytics, ERP systems	Real-time data processing, improved decision-making	High cost, complex integration
3	Brijraj Fashion: Style, Strength, and Success with Odoo by Odoo Team	2023	Case Study (Textile retail)	Odoo ERP	Reduced manual data entry, 50% time saving	Requires customization expertise
4	Industry 4.0 in Textile and	2024	Systematic Literature	VOS viewer, Scopus DB	Identifies major tech	No primary data or

	Apparel Industry: Systematic Review by B. Deepthi, Vikram Bansal		Review, Bibliometric Analysis		trends in smart textiles	industrial case
5	Odoo’s Transformation: Elevating Sportswear Manufacturing at TIE by Odoo Team	2023	Case Study (Manufacturing use-case)	Odoo ERP, Third-party integrations	Streamlined production, better analytics	Initial learning curve for employees
6	A Comprehensive Approach to Study the Adoption of Cloud ERP Among SMEs by Singh, Patel et al.	2023	UTAUT2 Model with Survey and Regression Analysis	Cloud ERP (Odoo, SAP B1), SPSS	Clarifies challenges and motivators for ERP adoption	May not apply to large enterprises
7	Use of Big Data in Textile Fashion and Garment Industry by N. Trader	2024	Conceptual Framework and Literature Study	Hadoop, Spark, BDA tools	Enhances forecasting, supply chain insights	Requires technical expertise, data strategy
8	Industrial Big Data Analytics: Challenges and Methodologies by JunPing Wang et al.	2018	Thematic Literature Analysis	IoT, Big Data Platforms	Scalable, data-driven decision-making	Generic, not specific to textile
9	Exploring Cloud ERP and Open Innovation for SMEs by Das, Akhtar et al.	2024	Qualitative Case Study, Interviews	Cloud ERP, Innovation Frameworks	Boosts agility, encourages collaboration	Responses subjective, not broadly applicable
10	BDA-Enabled Capabilities and Market Performance in Textile Sector	2024	Survey and Statistical Analysis (Textile firms in Pakistan)	BDA + ERP Tools, SPSS	Improved market responsiveness, strategy alignment	Limited to regional data (Pakistan)

	by Gulfam Haider, Ahmed et al.					

2.3 Hardware Requirements

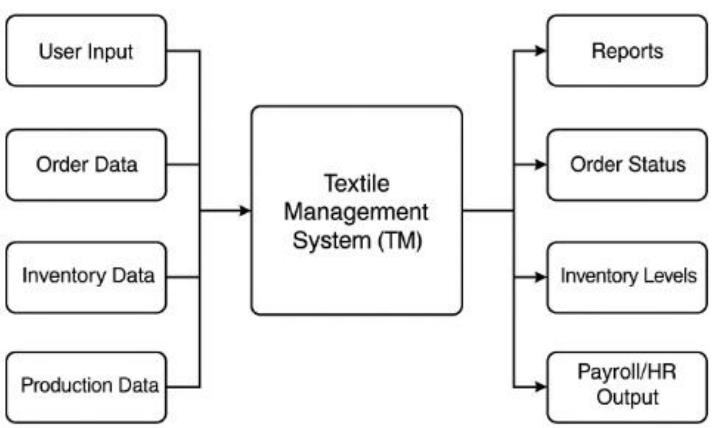
Component	**Specification**
Processor (CPU)	Intel Core i5 (10th Gen or newer) / AMD Ryzen 5 or higher
RAM	Minimum: 8 GB Recommended: 16 GB or higher for faster performance
Storage Drive	Minimum: 250 GB HDD Recommended: 512 GB SSD for quicker boot and response time
Motherboard	Compatible with Intel i5/Ryzen 5, supports DDR4 RAM, SATA/SSD connectors
Display Monitor	15” or larger screen Recommended: 21” Full HD (1920x1080 resolution)
Network Interface	Ethernet LAN (100 Mbps minimum) Wi-Fi (Dual-band preferred)
Graphics Card	Integrated graphics (sufficient for ERP use) Optional: NVIDIA GT 1030 or better
Power Supply Unit	450W or above, depending on system configuration
UPS (Power Backup)	600VA or higher (minimum 30 mins backup recommended)
Input Devices	Standard USB Keyboard and Optical Mouse
Peripheral Devices	Barcode Scanner (optional, for inventory tracking) Printer (optional)
Cabinet	ATX or Mini ATX cabinet with good airflow and USB 3.0 ports

2.4 Software Requirements

Software	Details
Operating System	Windows 10 / Linux Ubuntu 20.04 or higher
ERP Platform	Odoo Community Edition (v15 or later)
Database	PostgreSQL (v12 or later)
Programming Language	Python 3.8+ (used for Odoo customization)
Web Server	Nginx or Apache (for hosting on Linux-based systems)
Web Framework	Odoo's built-in MVC framework
Browser	Google Chrome / Mozilla Firefox
Development Environment	Visual Studio Code, PyCharm

3. System Design

3.1 High-Level Design (Black Box design)



1.Center Block: Represents the Textile Management System (TMS), which processes inputs and produces structured outputs.

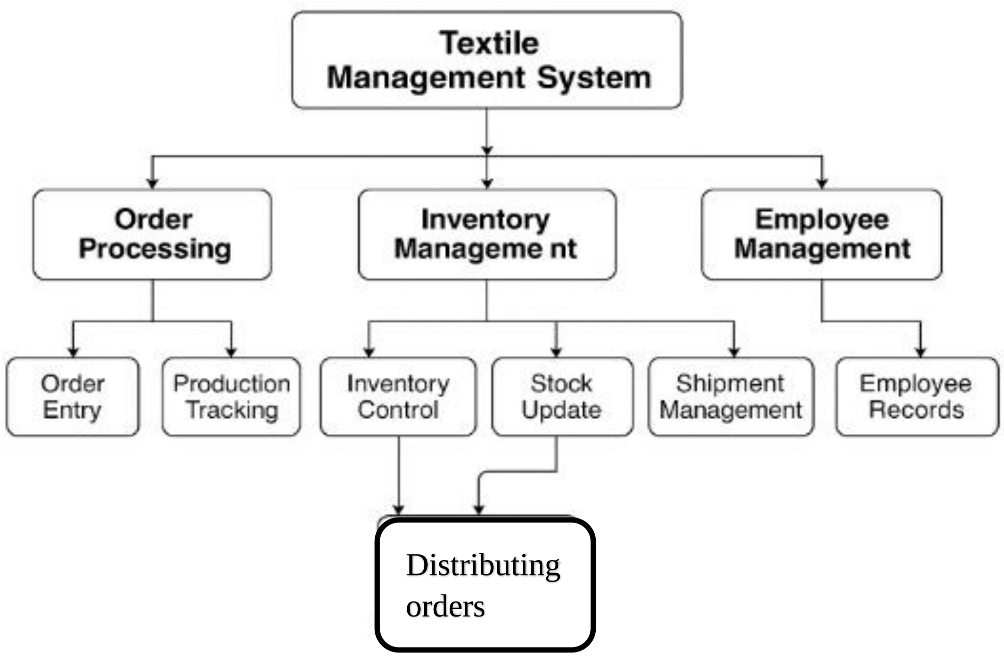
2.Left Side Inputs:

- **User Input** – Login, registration, and usage requests
- **Orders** – Customer and sales data
- **Inventory Data** – Material levels, stock entries
- **Production Data** – Machine and employee production status

3.Right Side Outputs:

- **Reports** – Sales, stock, HR, and production analytics
- **Order Status** – Real-time customer updates
- **Inventory Levels** – Current raw material and product stock
- **Payroll** – Salary calculations, HR summaries

3.2 Low-Level Design (Detailed design)



This diagram shows the Textile Management System (TMS) works internally. It is divided into three main modules: Order Processing, Inventory Management, and Employee Management—all controlled through the central TMS

1. Order Processing Module
Handles all customer orders.

- **Customer Details** – Stores customer info like name, contact, and address.
- **Order Entry** – Employees enter order details (product, size, quantity).
- **Billing & Invoicing** – Auto-generates bills and invoices.
- **Order Tracking** – Shows current status of the order

2. Inventory Management Module
Manages stock and materials.

- **Stock Entry** – Records new raw materials or products.
- **Material Usage** – Tracks material use in production.
- **Reorder Alerts** – Notifies when stock is low.
- **Warehouse Mapping** – Shows where items are stored.

3. Employee Management Module
Takes care of staff-related functions.

- **Staff Records** – Stores employee details.
- **Attendance** – Tracks daily attendance.
- **Payroll** – Calculates salaries automatically.
- **Performance Review** – Monitors employee performance.

4.distributing orders

- Delivery of orders to customers

4. System Implementation
4.1 Algorithms (followed, proposed or altered)

- **Order Processing Algorithm**
  Goal: To register customer orders and update order status.
  Steps:
  1. Input customer details and order specifications.
  2. Validate product availability (check inventory).
  3. If available, create an order and assign a unique ID.
  4. Generate invoice and set status to "Processing."
  5. Update status to "Shipped" after dispatch.

  Result: Reduces manual entry errors and speeds up processing.

- **Inventory Management Algorithm**

  Goal: To update stock based on material usage and order fulfilment.

  Steps:
  1. On order placement, subtract used items from stock.
  2. If stock level < reorder threshold, generate alert.
  3. On new stock arrival, update inventory database.
  4. Maintain logs for audit purposes.

  Result: Prevents stock-out and overstocking situations.

- **Payroll Calculation Algorithm**

  Goal: To calculate monthly salary of employees.

  Steps:
  1. Fetch attendance and shift logs for each employee.
  2. Compute total working hours and overtime.
  3. Applysalaryformula:
     Salary = (Base Salary + Overtime) - Deductions
  4. Generate payslip and store in records.

  Result: Ensures fair and transparent salary calculation.

4.2 Module Development –Code

1.Order Processing (Odoo Python Controller)

```
class Order(models.Model):
    _name = 'tms.order'
customer_name = fields.Char(string='Customer Name')
    product = fields.Char(string='Product')
    quantity = fields.Integer(string='Quantity')
    status = fields.Selection([
        ('draft', 'Draft'),
        ('processing', 'Processing'),
        ('shipped', 'Shipped')
    ], default='draft')

    def process_order(self):
        self.status = 'processing'
def ship_order(self):
        self.status = 'shipped'
```

2.Inventory Management (Stock Update Logic)

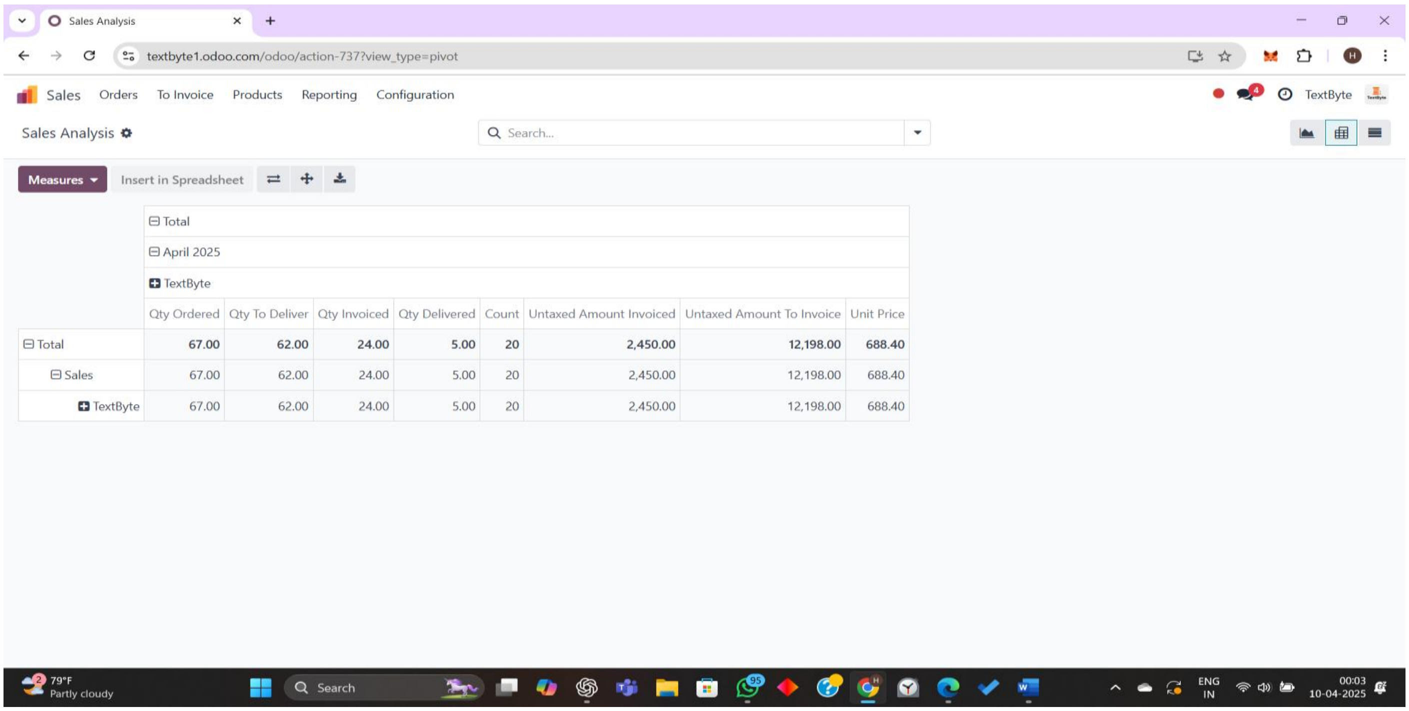
```
class Inventory(models.Model):
    _name = 'tms.inventory'

    product_name = fields.Char(string='Product Name')
    quantity_available = fields.Integer(string='Available Stock')
    reorder_level = fields.Integer(string='Reorder Threshold')

    def update_stock(self, used_qty):
        self.quantity_available -= used_qty
        if self.quantity_available <= self.reorder_level:
            self.send_reorder_alert()

    def send_reorder_alert(self):
        # Notify purchase team
        print(f"Reorder needed for: {self.product_name}")
```

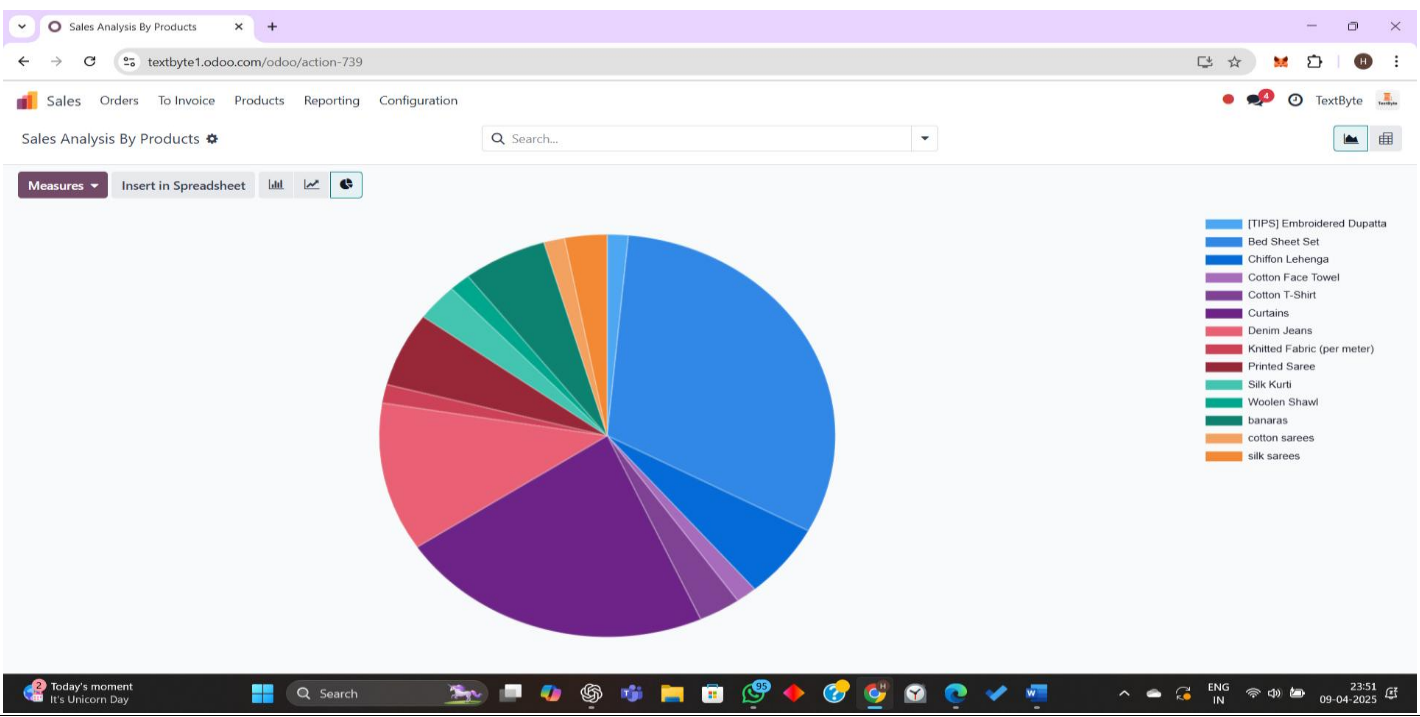
3. **Payroll Calculation**

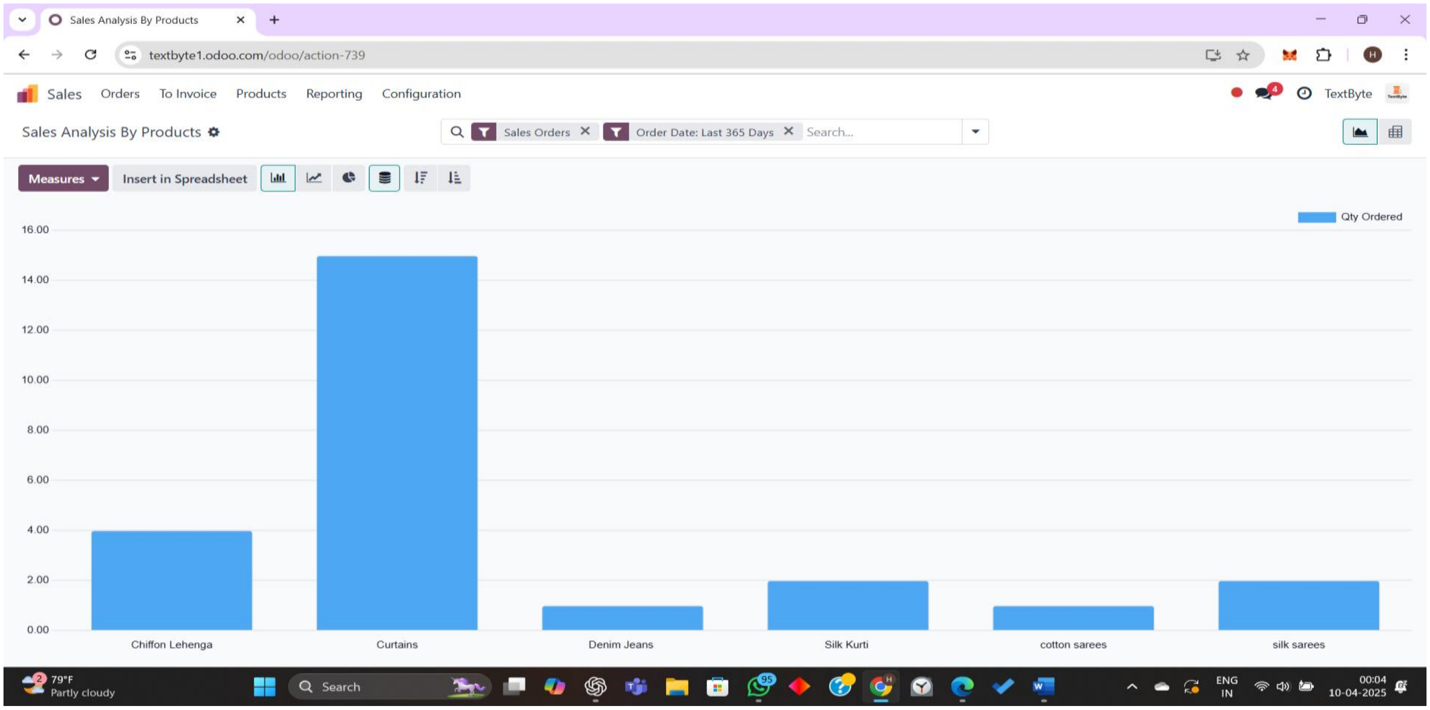
```
class Payroll(models.Model):
    _name = 'tms.payroll'

    employee = fields.Many2one('hr.employee')
    base_salary = fields.Float(string='Base Salary')
    overtime_hours = fields.Float(string='Overtime Hours')
    deductions = fields.Float(string='Deductions')

    @api.depends('overtime_hours', 'deductions')
    def compute_salary(self):
        for record in self:
            overtime_pay = record.overtime_hours * 100  # ₹100/hour
            record.total_salary = record.base_salary + overtime_pay - record.deductions
```

5.Results and Discussion

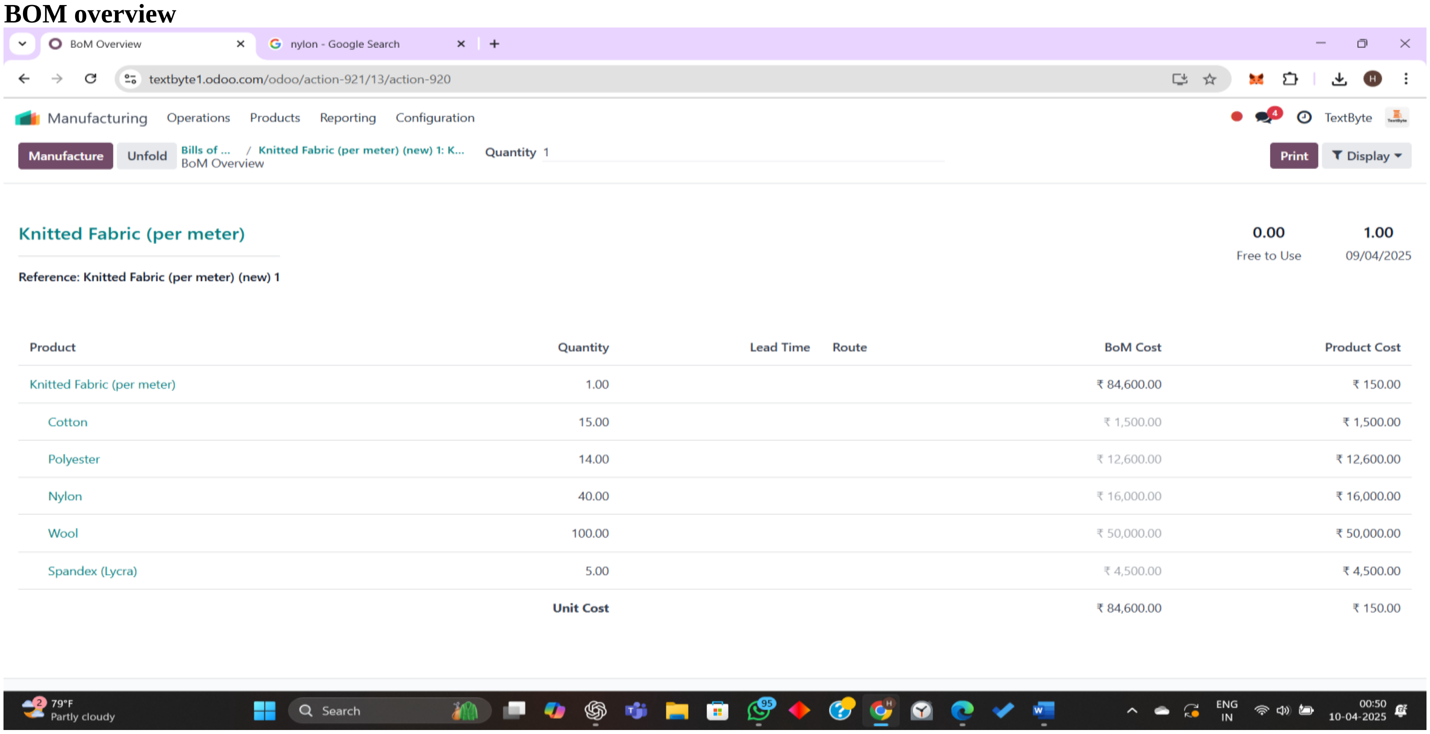
5.1 Implementation Results

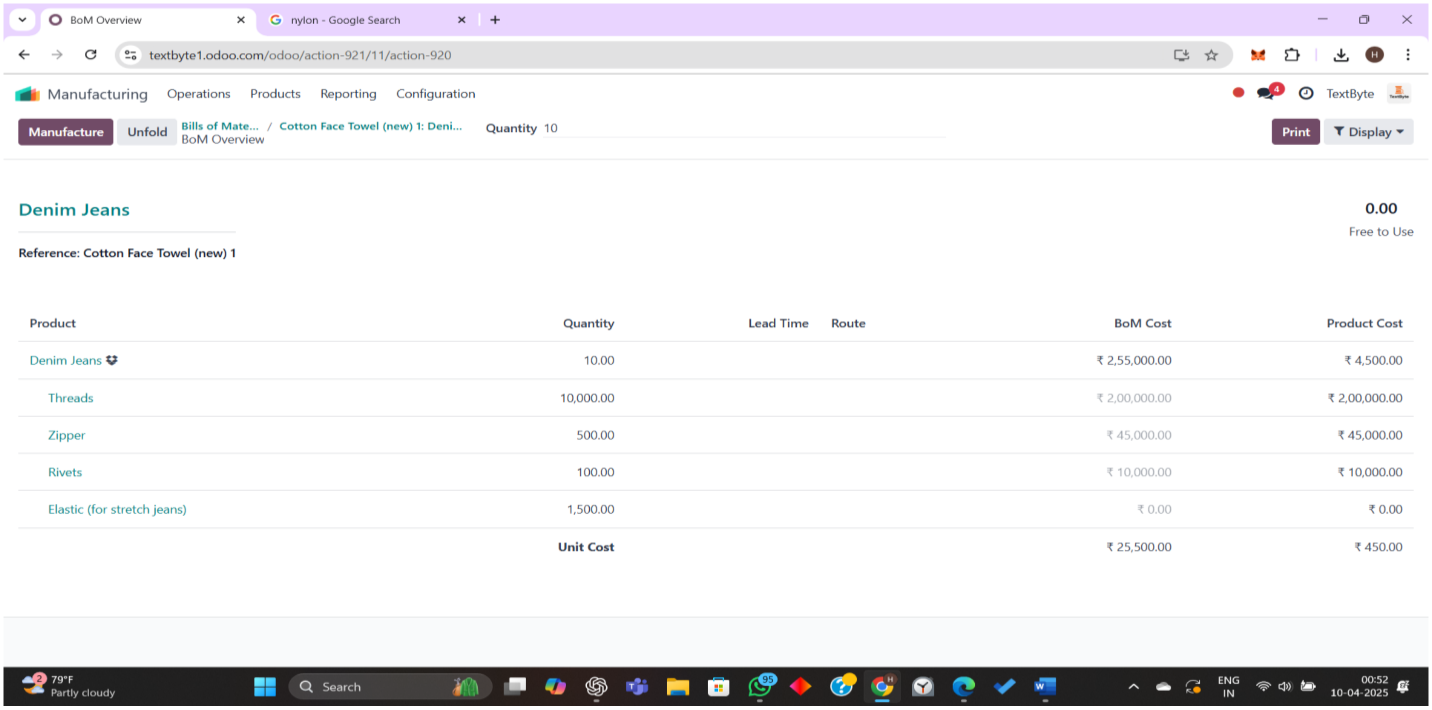




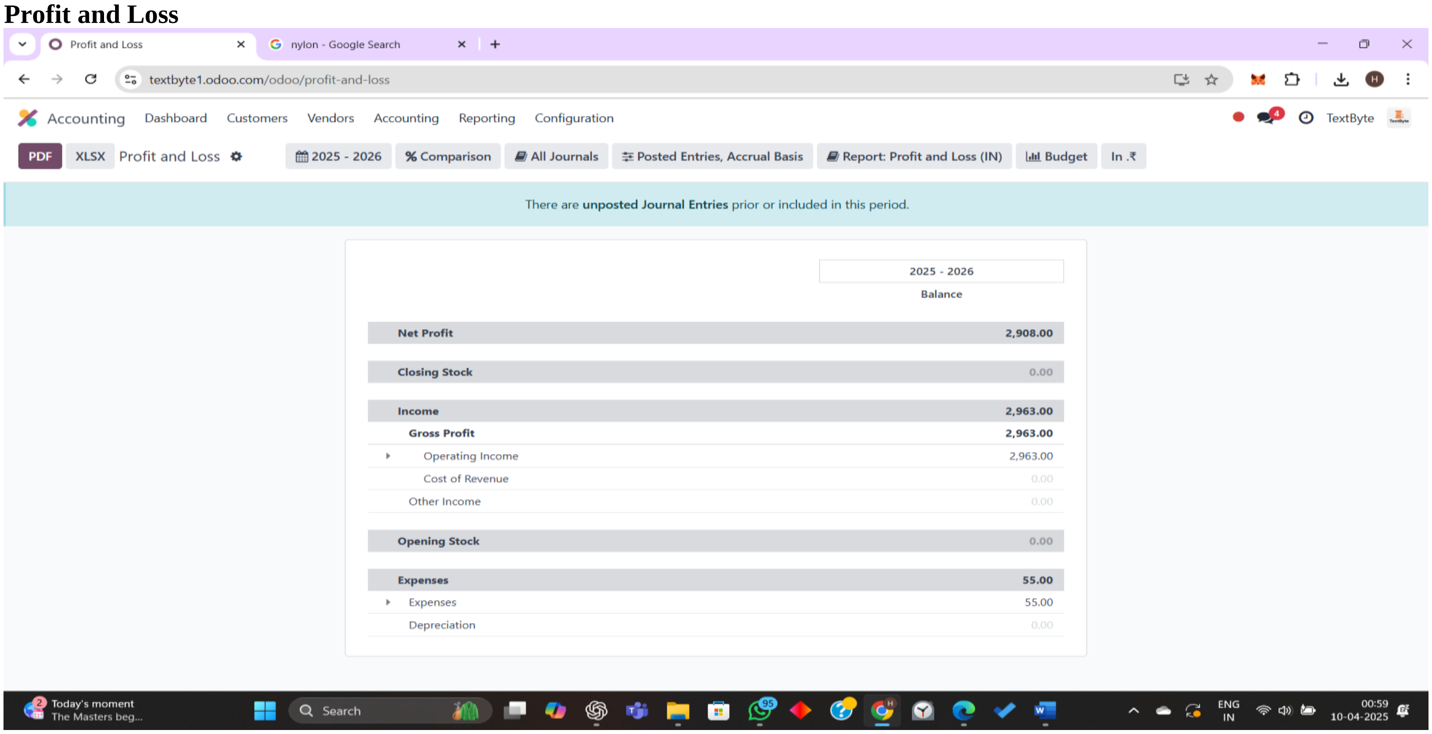


BOM overview





Profit and Loss



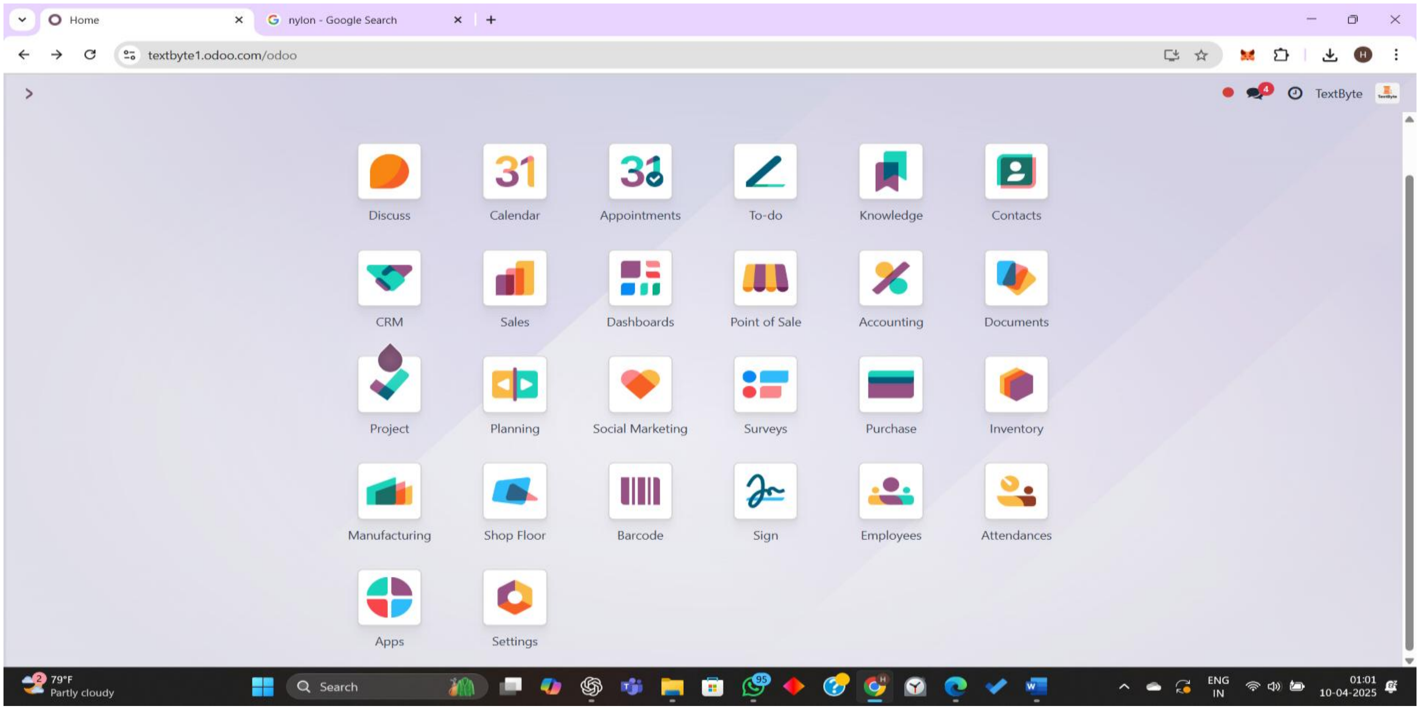
5.2 Metrics

The performance and effectiveness of the Integrated Beverage Selling System can be evaluated using the following metrics:

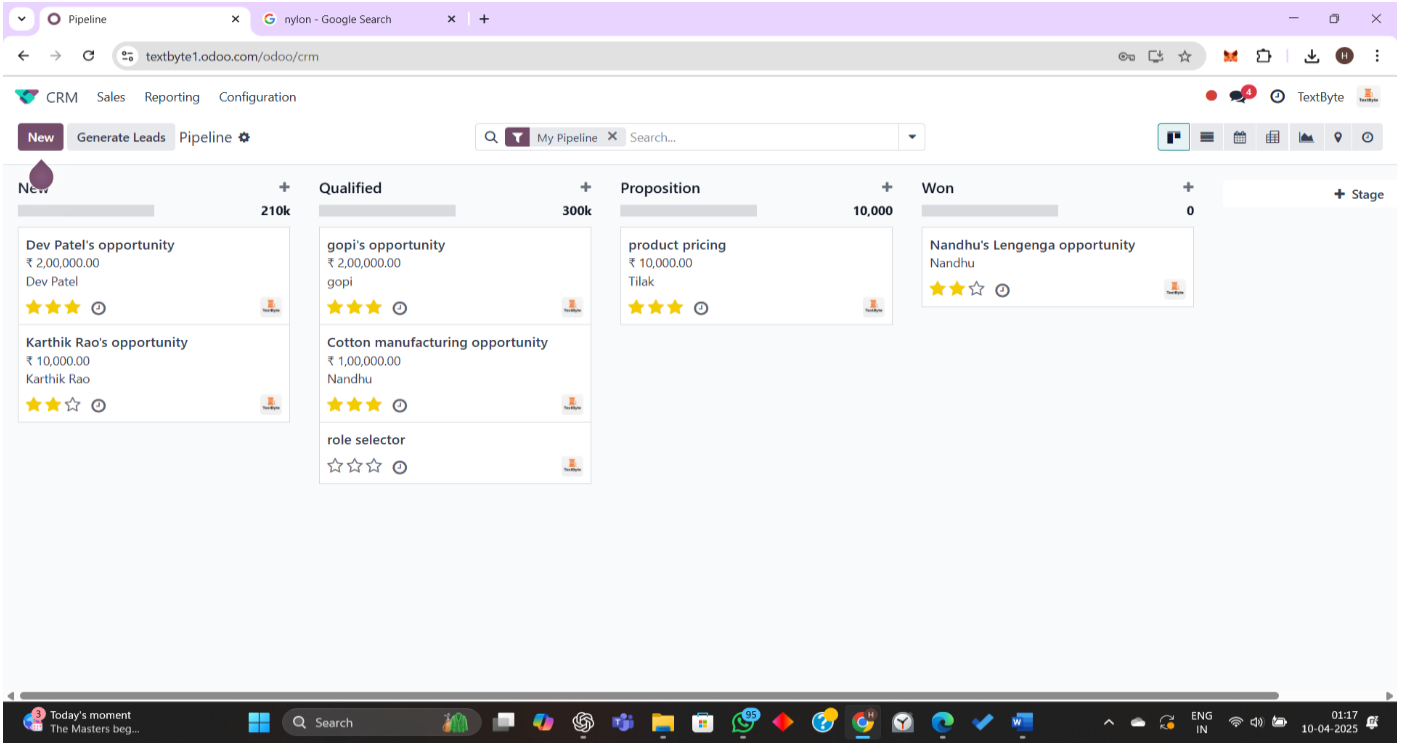
- **Order Processing Time:** Measures the duration from order placement to fulfillment to ensure efficiency.
- **Inventory Accuracy Rate**: Tracks discrepancies between recorded and actual stock levels to minimize errors.
- **Customer Satisfaction Score (CSAT):** Captures user feedback on their experience, providing insights into service quality.
- **Sales Conversion Rate:** Determines the percentage of visitors who complete a purchase, indicating the effectiveness of the system’s interface and marketing strategies.
- **Delivery Efficiency:** Compares the actual delivery time against estimated schedules to assess logistics performance.
- **System Downtime:** Monitors system availability and reliability.
- **ERP Integration Success Rate:** Evaluates how effectively transactions are processed through the Odoo ERP system.
- **Repeat Purchase Rate:** Indicates customer loyalty by tracking the percentage of users making multiple purchases over time.
- **AI Recommendation Effectiveness:** Measures how well AI-based product suggestions influence customer decisions.
- **Customer Retention Rate:** Assesses the ability to maintain long-term customer relationships and recurring sales.

5.3 Screen Shots

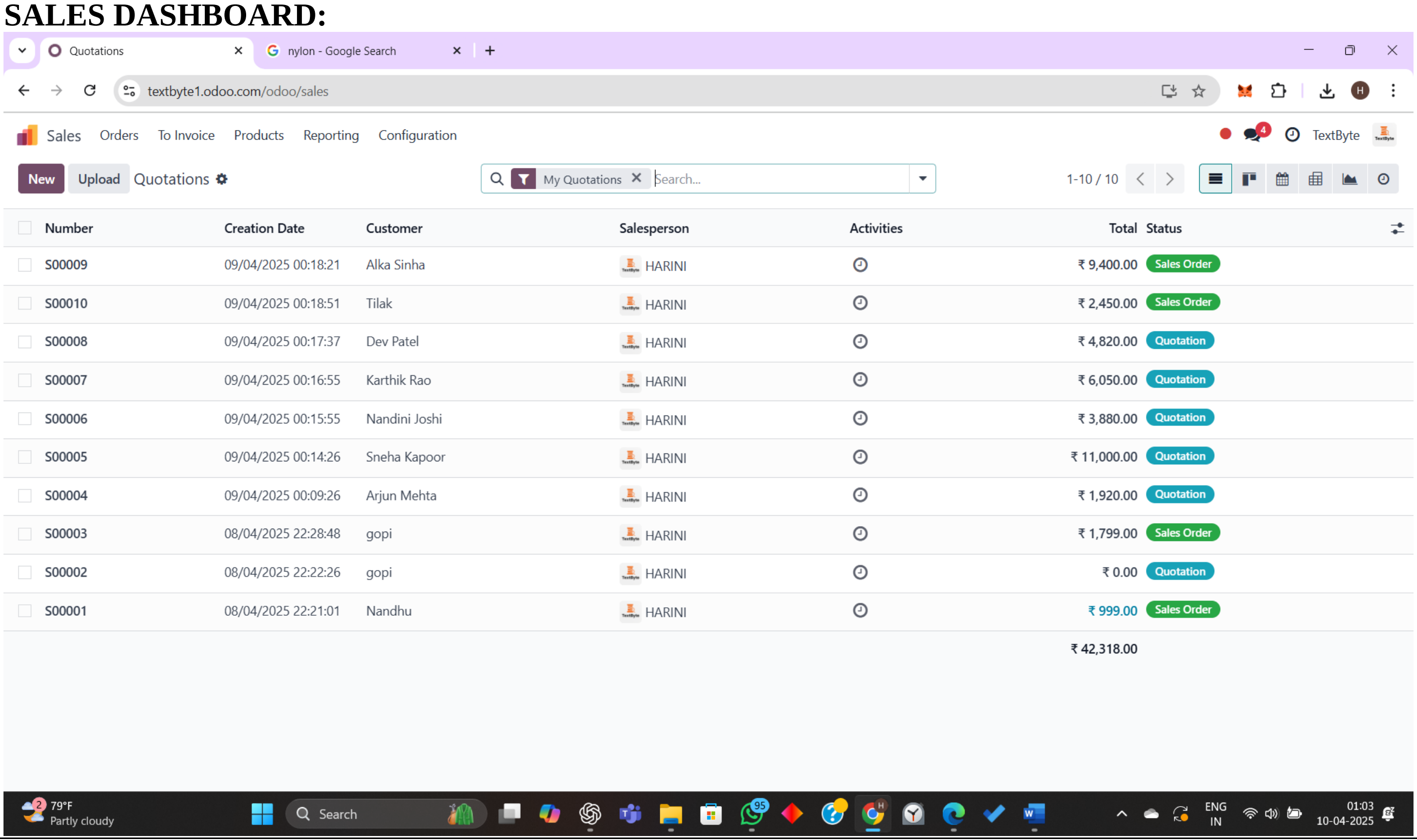
ODOO MODULES:



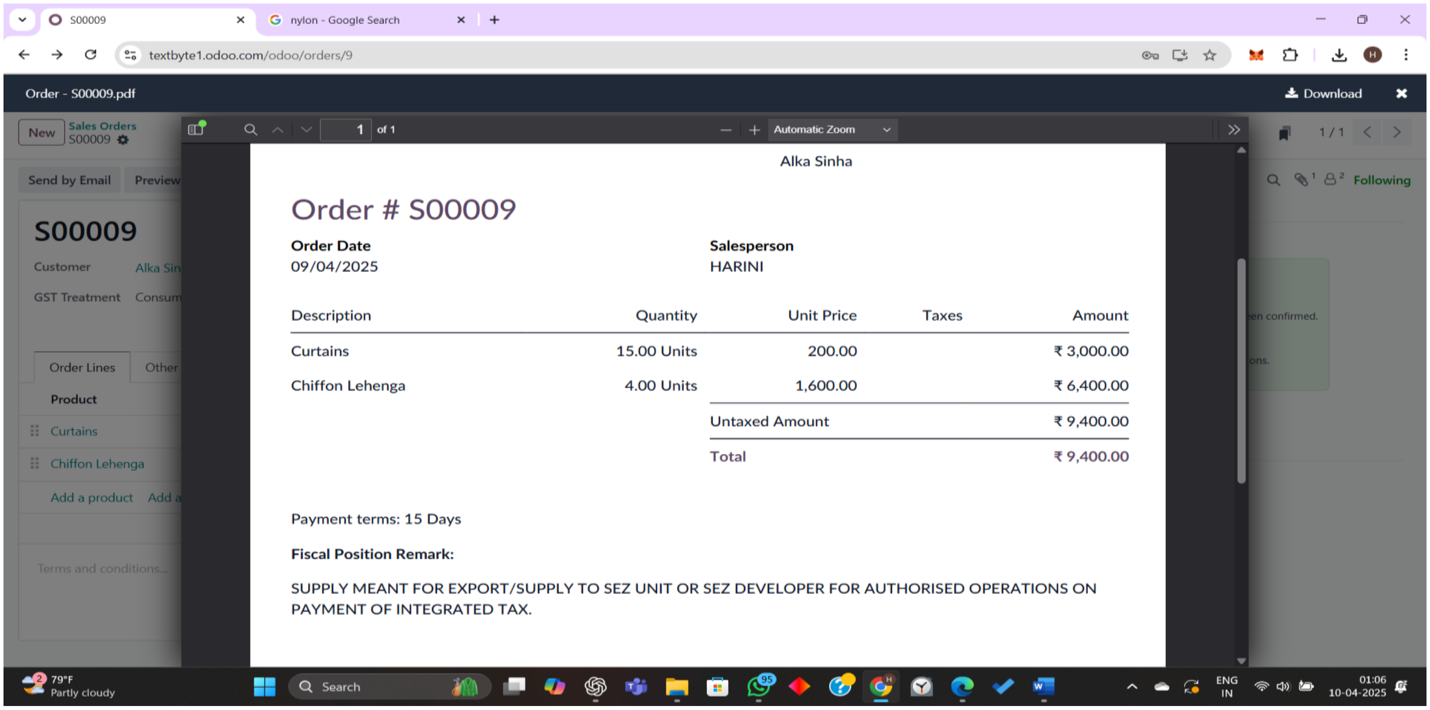
CRM:



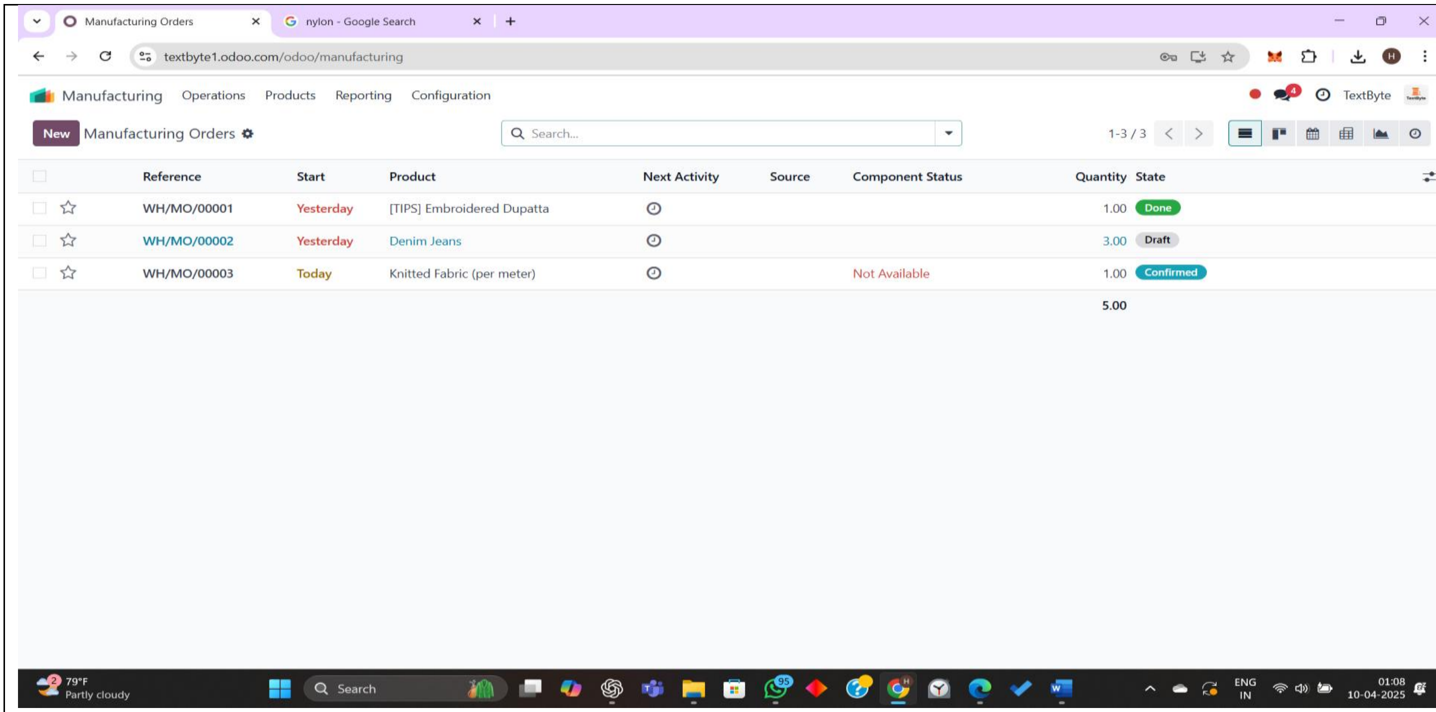
SALES DASHBOARD:



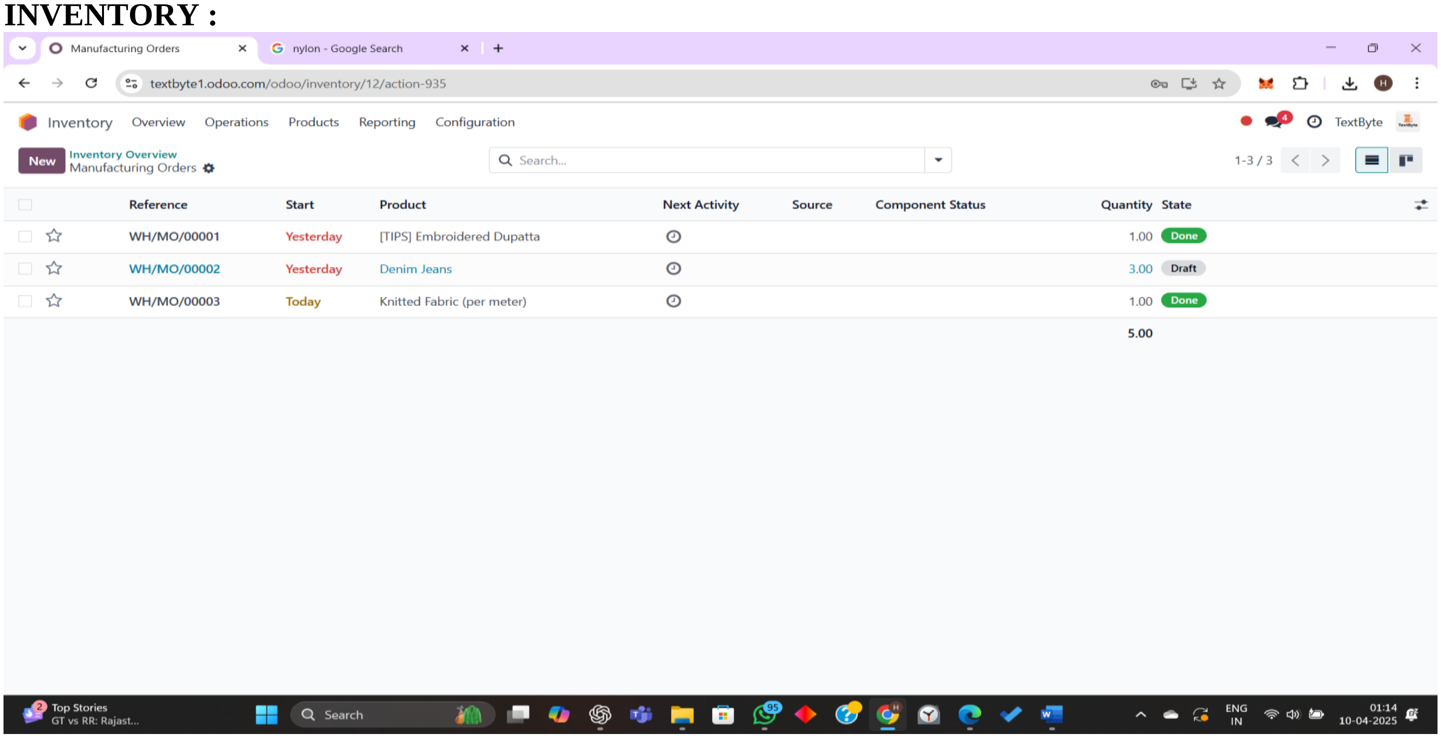
ORDER DASHBOARD:



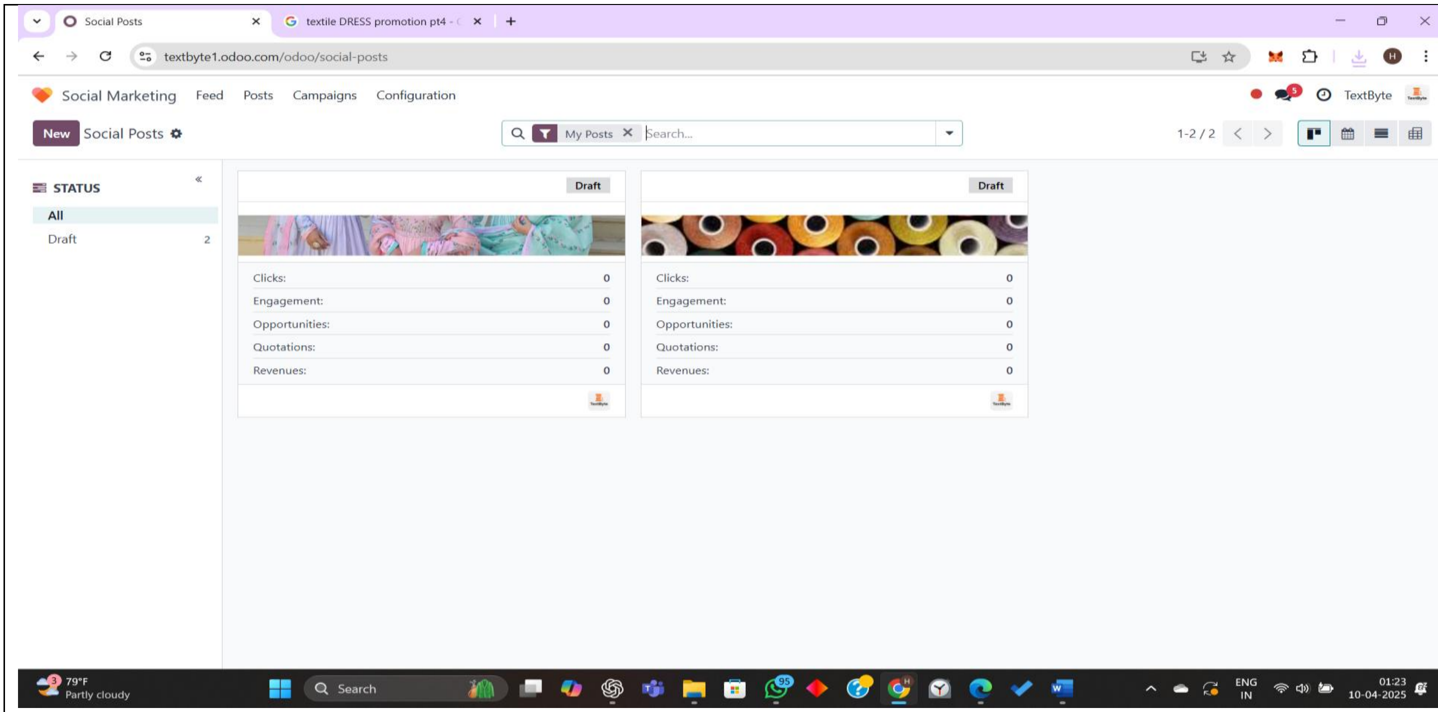
MANUFACTURING :



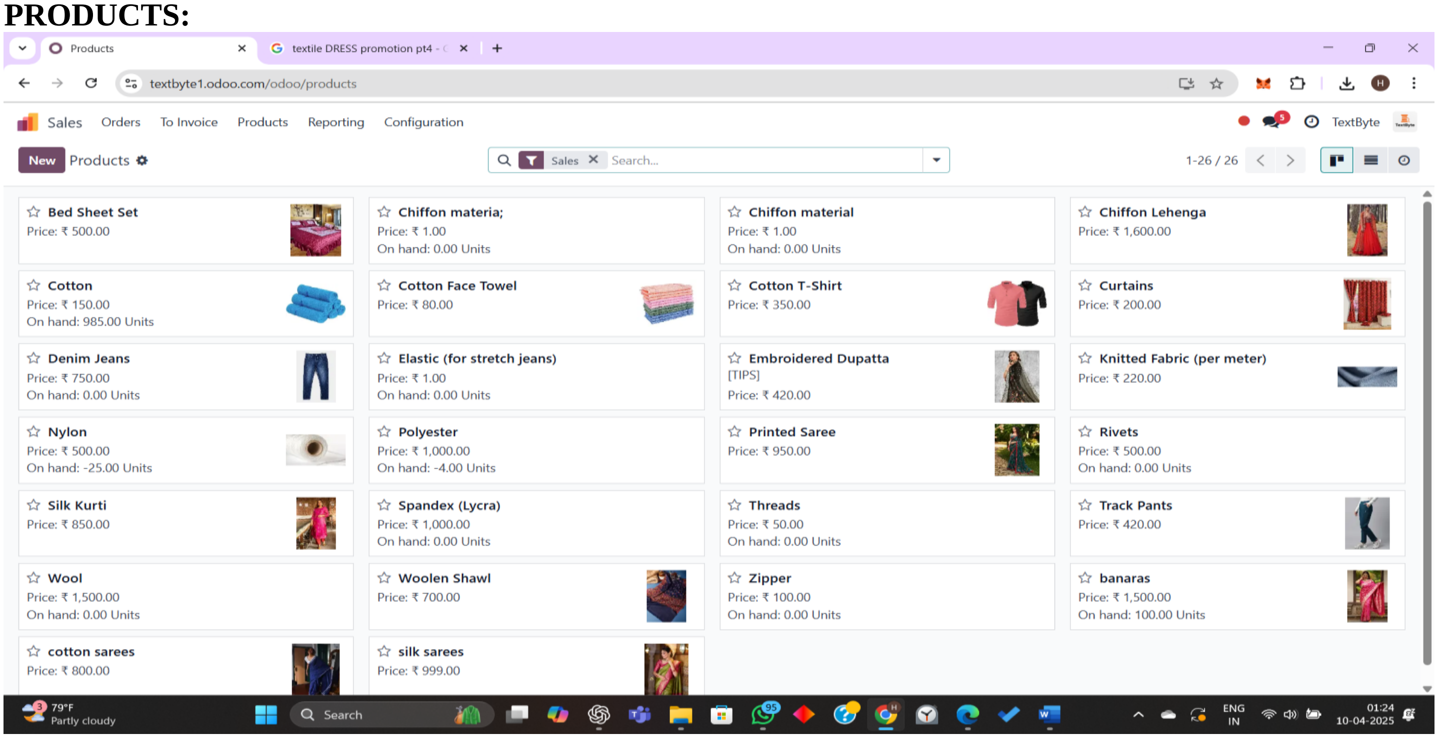
INVENTORY :



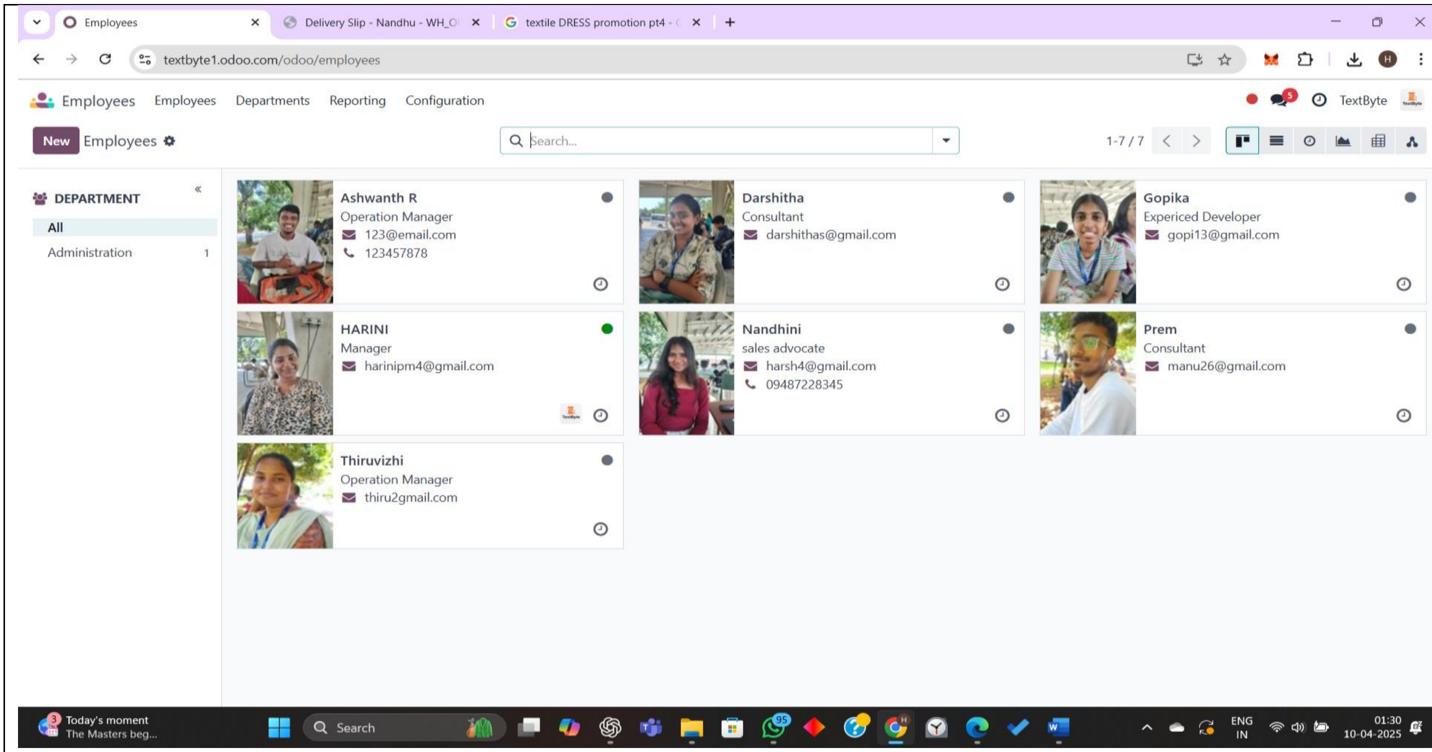
SOCIAL MARKETING:



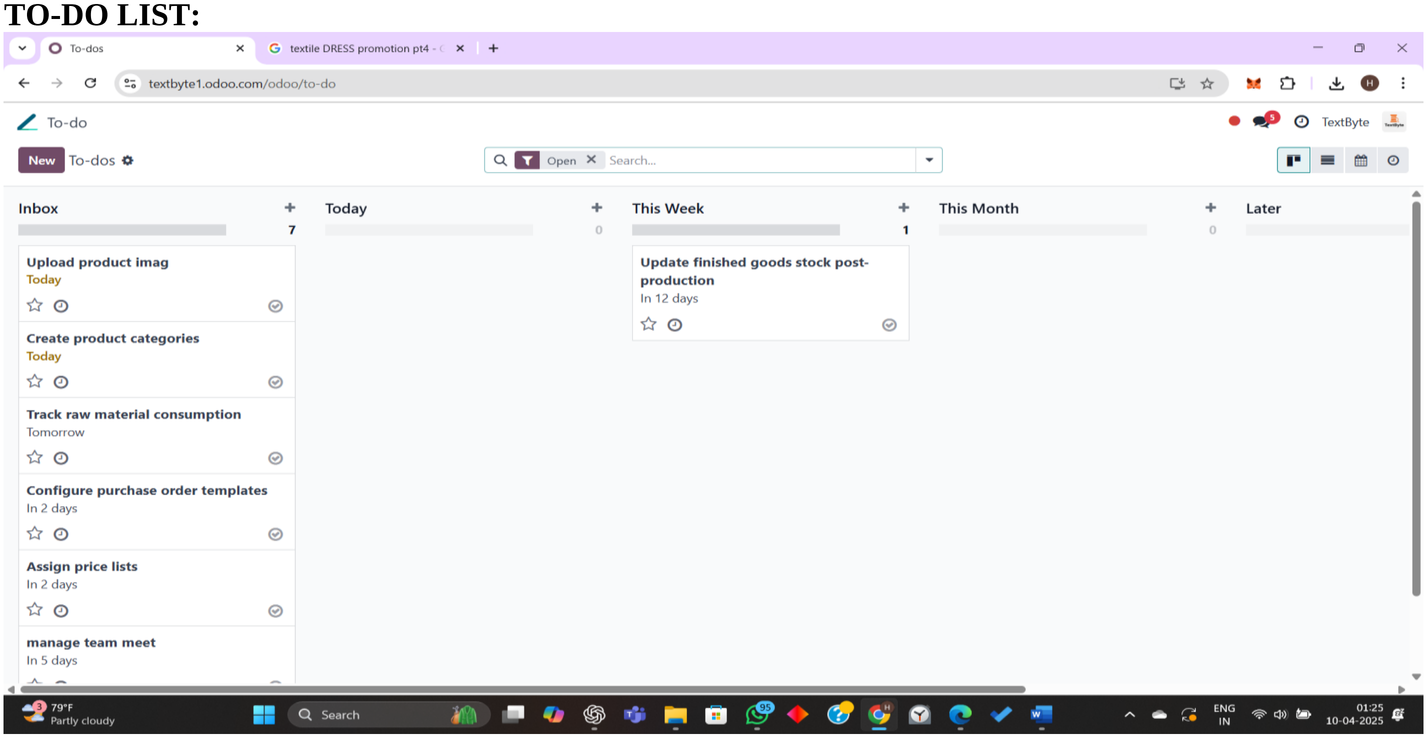
PRODUCTS:



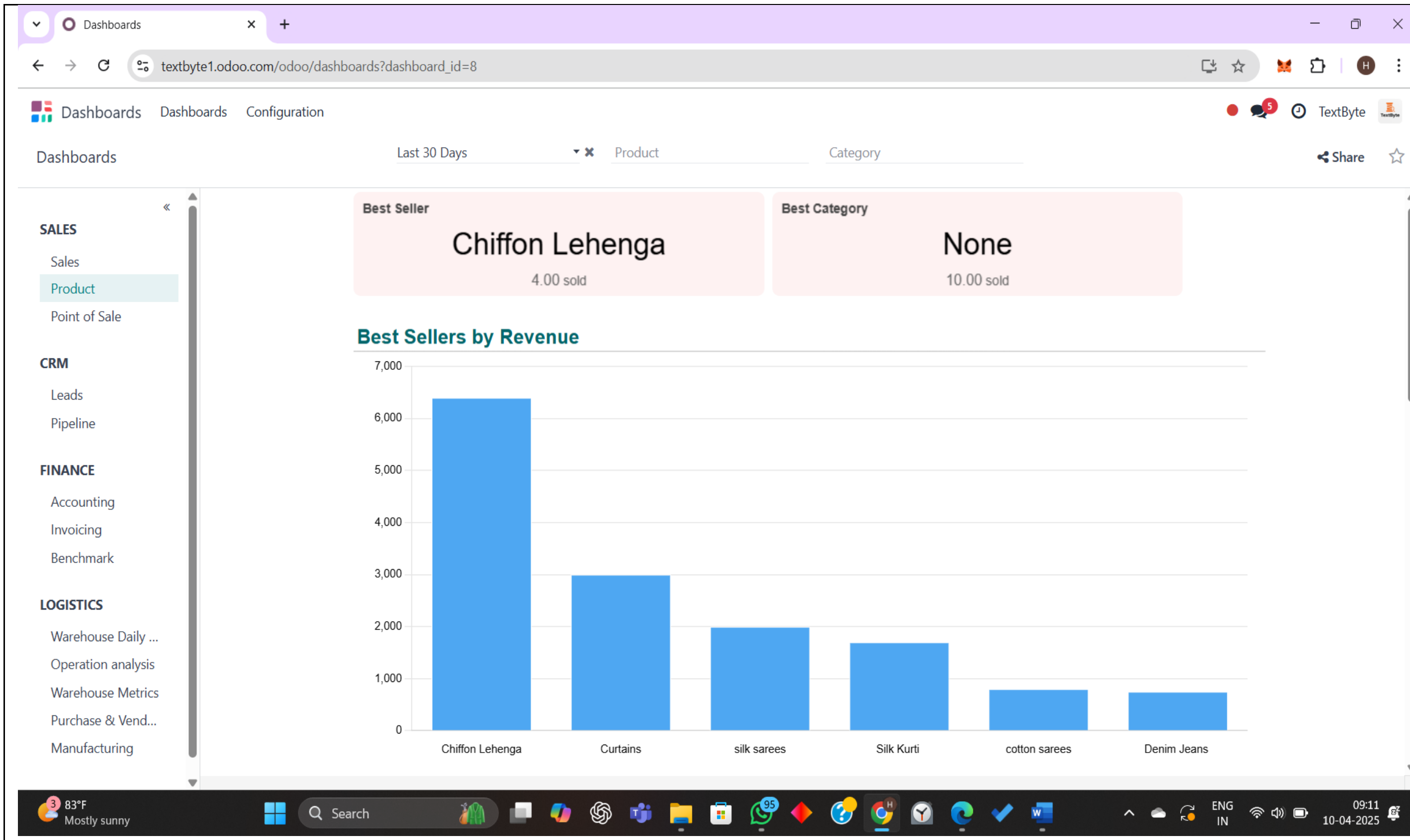
EMPLOYEES:



TO-DO LIST:



PURCHASEs: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Pipeline
FINANCE
Accounting
Invoicing
Benchmark
LOGISTICS
Warehouse Daily ...
Operation analysis
Warehouse Metrics
Purchase & Vend...
Manufacturing
Best Seller
Chiffon Lehenga
4.00 sold
Best Category
None
10.00 sold
Best Sellers by Revenue
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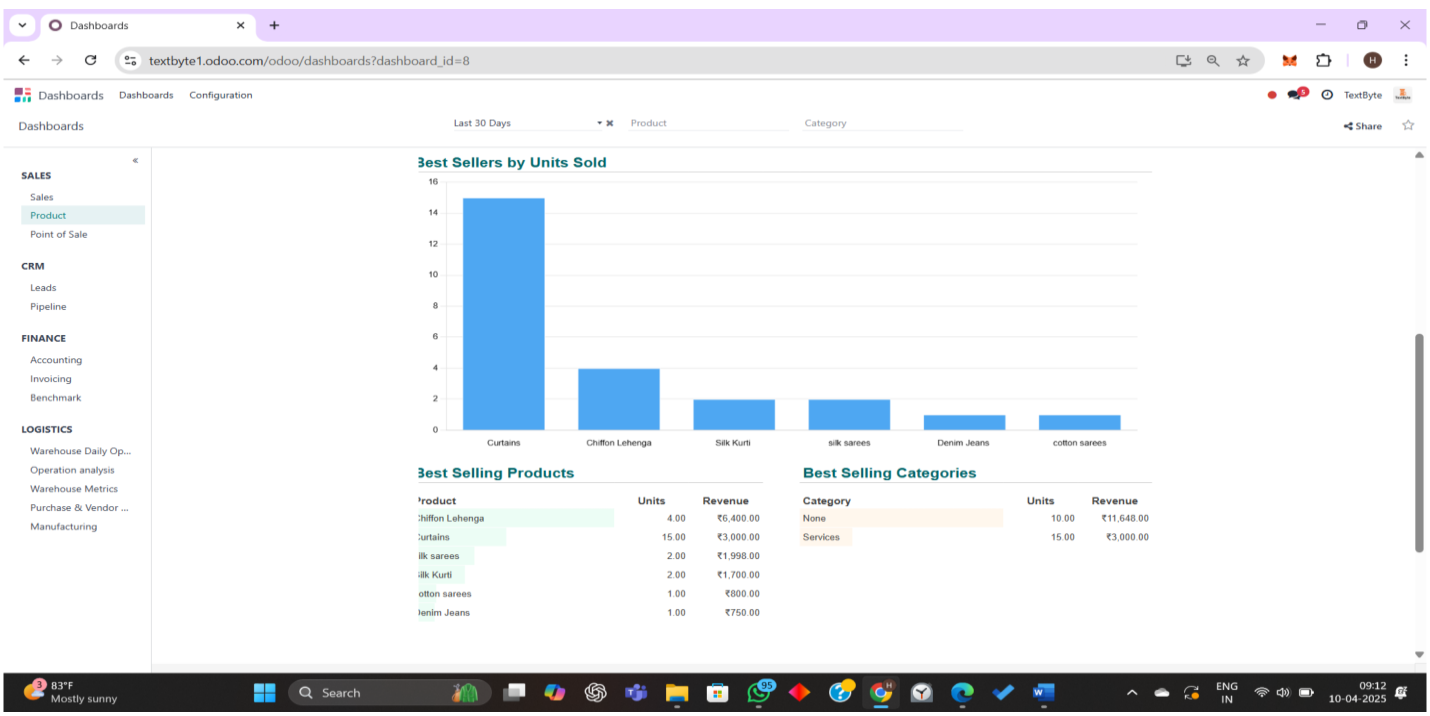
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Dashboards
Last 30 Days
Product
Category
Share
SALES
Sales
Product
Point of Sale
CRM
Leads
Pipeline
FINANCE
Accounting
Invoicing
Benchmark
LOGISTICS
Warehouse Daily Op...
Operation analysis
Warehouse Metrics
Purchase & Vendor ...
Manufacturing
Best Sellers by Units Sold
16
14
12
10
8
6
4
2
0
Curtains
Chiffon Lehenga
Silk Kurti
silk sarees
Denim Jeans
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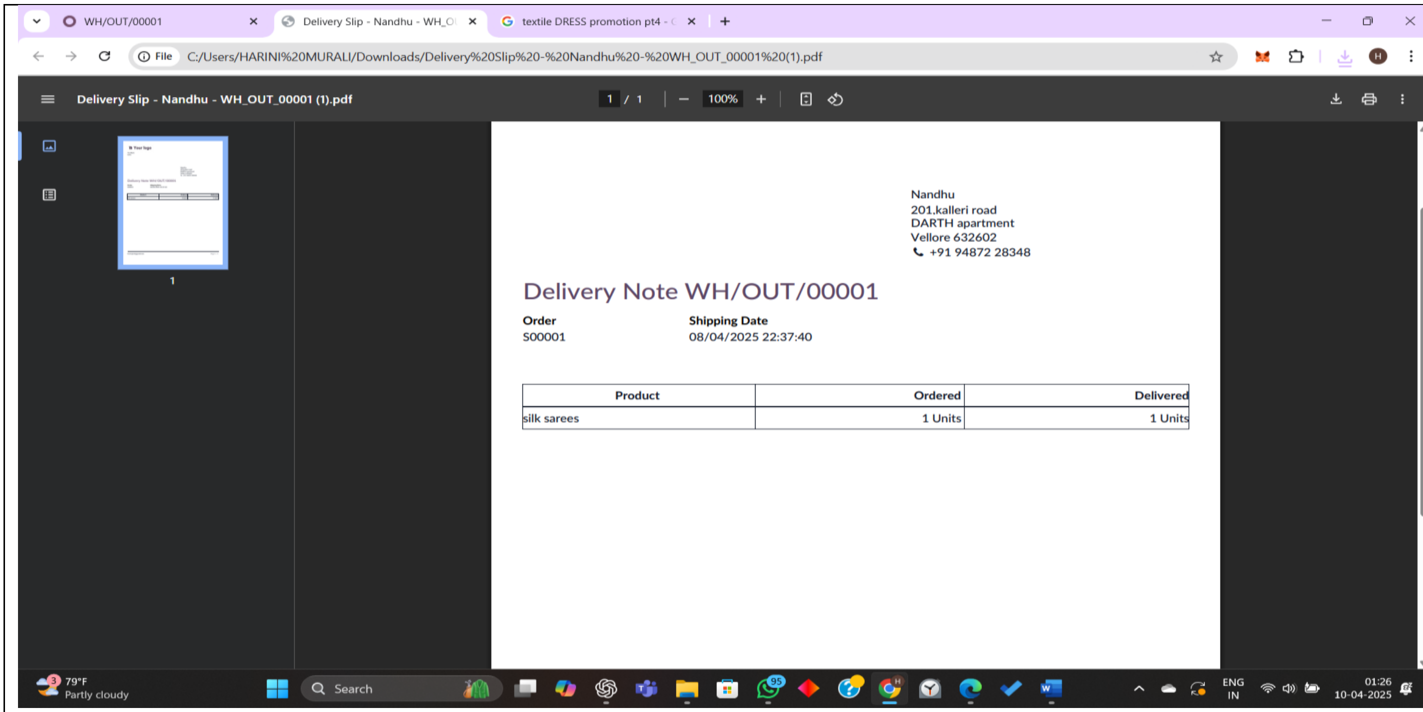

5.4 Mapping the results with problem statement and existing systems

Problem Area	**Problem Faced**	**Solution by TMS**	**Result Achieved**
Order Management	Manual entry and status tracking	Automated order entry, billing, and live tracking	Faster processing, accurate billing, and customer updates
Inventory Management	Stockouts and over-purchasing	Real-time stock updates and reorder alerts	Reduced stock wastage and smoother procurement
Employee Attendance & Payroll	Manual salary and attendance tracking	Auto-calculated salary based on digital attendance logs	Transparent and error-free payroll
Report Generation	Time-consuming and inconsistent reporting	Auto-generated reports for sales, inventory, and HR	Quick decision-making and improved monitoring

Lack of Centralized Data	Data scattered across departments	Unified ERP platform with role-based access	Increased efficiency and coordination

Comparison with Existing System

Feature	Traditional System	Proposed TMS (Odoo-based)
Order Handling	Manual entry in registers or Excel	Digital forms with workflow automation
Stock Control	Manual logs; error-prone	Real-time stock tracking
Salary Calculation	Manual or Excel based	Auto-calculated, system-driven
Reporting	Delayed and inconsistent	Real-time, visual reports
Accessibility	Local-only, paper-based	Web-based access from anywhere

5.5 Discussions

The Textile Management System (TMS) was developed to solve the common issues faced in the textile industry, such as manual errors, delays, and poor coordination. After implementation, the system showed positive results:

- Order processing became faster and more organized.
- Inventory updates were real-time and more accurate.
- Employee data and payroll were managed automatically, reducing errors.

Key Points:

- The system is easy to use and requires little training.
- Integration between modules (orders, inventory, HR) worked smoothly.
- Reports helped managers make quicker decisions.

Challenges:

- Attendance and payroll module needed some extra customization.
- Managing simultaneous stock updates needed careful handling.

Overall, the project proved that using an ERP-based system like TMS can greatly improve efficiency and accuracy in textile operations.

6. Conclusion and Future Developments

Conclusion:

The implementation of the Textile Management System (TMS) successfully addresses many of the inefficiencies present in traditional textile industry operations. By automating key processes such as order management, inventory control, and employee tracking, the system significantly reduces manual workload, increases data accuracy, and improves overall operational efficiency.

Using an open-source ERP platform like Odoo, the system integrates various departments into a unified platform, providing real-time visibility and decision support. The modular design ensures that the system remains user-friendly, scalable, and adaptable to a range of textile business needs.

Future Developments:

While the current version of the system offers strong foundational features, there is potential for further enhancement:

1. **Mobile App Integration:**

   Allow users and employees to access key functionalities on the go (e.g., order tracking, leave application).

2. **AI-Based Demand Forecasting**:

   Implement machine learning models to predict customer demand and automate procurement planning.

3. **Barcode and RFID Integration**

   Use advanced tracking for inventory and product movement within the warehouse.

4. **Customer Portal**

   Provide customers with a login area to place orders, track status, and download invoices.

5. **Multilingual & Multi-Currency Support**

   Improve accessibility for international customers and suppliers.

6. **Cloud Deployment**

   Host the system on the cloud for better scalability, accessibility, and data backup.

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